Programming languages Java

Polymorphism

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Polymorphism overview

Polymorphism: using a type/method/object in different ways

Categories

- universal polymorphism (applicable in *infinitely many*, extensible ways)
  - ◇ parametric polymorphism: generics
  - ◇ subtype polymorphism: inheritance
- ad hoc polymorphism (applicable in specific, *finite* number of ways)
  - ◇ overloading
  - ◇ casting: explicit type conversion

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An earlier example

```
public class Receptionist {
  public Time[] readWakeupTimes(String[] fnames) {
    Time[] times = new Time[fnames.length];

    for (int i = 0; i < fnames.length; ++i) {
      try {
        times[i] = readTime(fnames[i]);
      } catch (java.io.IOException e) {
        times[i] = null;   // no-op
        System.err.println("Could not read " + fnames[i]);
      }
    }
    return times; // maybe sort times before returning?
  }
  ...
}
```

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Getting rid of null values

```
public class Receptionist {
  public Time[] readWakeupTimes(String[] fnames) {
    Time[] times = new Time[fnames.length];
    int len = 0;
    for (int i = 0; i < fnames.length; ++i) {
      try {
        times[len] = readTime(fnames[i]);
        ++len;
      } catch (java.io.IOException e) {
        System.err.println("Could not read " + fnames[i]);
      }
    }
    return java.util.Arrays.copyOf(times,len); // possibly
                                               // sorted
  }
  ...
}
```

Advantages and drawbacks of arrays

- Efficient access to elements (indexing)
- Syntactic support in the language (indexing, array literals)
- Length is fixed at construction
  - ◇ Extension with, or removal of, elements is expensive
  - ◇ Requires the creation of a new array, and copying
- Some methods have unexpected (wrong) implementations
  - ◇ This makes arrays incompatible with some data structures

Alternative: java.util.ArrayList

convenient standard library, similar inner workings

```
String[] names = { "Tim",
                   "Jerry" };


names[0] = "Tom";
String mouse = names[1];

String[] trio = new String[3];
trio[0] = names[0];
trio[1] = names[1];
trio[2] = "Spike";
names = trio;
```

Alternative: java.util.ArrayList

convenient standard library, similar inner workings

```
String[] names = { "Tim",
                   "Jerry" };


names[0] = "Tom";
String mouse = names[1];

String[] trio = new String[3];
trio[0] = names[0];
trio[1] = names[1];
trio[2] = "Spike";
names = trio;
```

```
ArrayList<String> names =
          new ArrayList<>();
names.add("Tim");
names.add("Jerry");

names.set(0, "Tom");
String mouse = names.get(1);

names.add("Spike");
```

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Example updated

```
public class Receptionist {
  ...
  public ArrayList<Time> readWakeupTimes(String[] fnames) {
    ArrayList<Time> times = new ArrayList<Time>();
    for (int i = 0; i < fnames.length; ++i) {
      try {
          times.add(readTime(fnames[i]));
      } catch (java.io.IOException e) {
          System.err.println("Could not read " + fnames[i]);
      }
    }
    return times; // possibly sort before returning
  }
}
```

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Parametrized type

```
ArrayList<Time> times
```

```
Time[] times
Time times[]
```

Generic class

Not exactly this way, but almost···

```
package java.util;
public class ArrayList<T> {
    public ArrayList() { ... }
    public T get(int index) { ... }
    public void set(int index, T item) { ... }
    public void add(T item) { ... }
    public T remove(int index) { ... }
    ...
}
```

Type parameter is provided when used

```
import java.util.ArrayList;

...

ArrayList<Time> times;
ArrayList<String> names = new ArrayList<String>();
ArrayList<String> namez = new ArrayList<>();
```

Generic method

```
import java.util.*;
public class Main {
  public static <T> void reverse(T[] array) {
    int lo = 0, hi = array.length-1;
    while (lo < hi) {
      T tmp = array[hi];
      array[hi] = array[lo];
      array[lo] = tmp;
      ++lo; --hi;
    }
  }
  public static void main(String[] args) {
    reverse(args);
    System.out.println(Arrays.toString(args));
  }
}
```

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Generic method with manually specified type argument

```
import java.util.*;
public class Main {
  public static <T> void reverse(T[] array) {
    int lo = 0, hi = array.length-1;
    while (lo < hi) {
      T tmp = array[hi];
      array[hi] = array[lo];
      array[lo] = tmp;
      ++lo; --hi;
    }
  }
  public static void main(String[] args) {
    Main.<String>reverse(args);
    System.out.println(Arrays.toString(args));
  }
}
```

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Parametric polymorphism

- The same code works for many type parameters
  - ◇ What sort of code can take type parameters?
    - types (classes)
    - methods
- Code parametrized with any (reference) type

Type parameter

Primitive types not allowed!

```
ArrayList<int> numbers     // compilation error
```

Type parameter

Primitive types not allowed!

```
ArrayList<int> numbers     // compilation error
```

Reference types are OK!

```
ArrayList<Integer> numbers = new ArrayList<>();
numbers.add(Integer.valueOf(7));
Integer seven = numbers.get(0);
```

Type parameter

Primitive types not allowed!

```
ArrayList<int> numbers     // compilation error
```

Reference types are OK!

```
ArrayList<Integer> numbers = new ArrayList<>();
numbers.add(Integer.valueOf(7));
Integer seven = numbers.get(0);
```

Automatic conversion from/to primitive values

```
ArrayList<Integer> numbers = new ArrayList<>();
numbers.add(42);              // auto-boxing:   int -> Integer
int n42 = numbers.get(1); // auto-unboxing: Integer -> int
```

Data structures in java.util

Sequence

```
ArrayList<String> colors = new ArrayList<>();
colors.add("red"); colors.add("white"); colors.add("red");
String third = colors.get(2);
```

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Data structures in java.util

Sequence

```
ArrayList<String> colors = new ArrayList<>();
colors.add("red"); colors.add("white"); colors.add("red");
String third = colors.get(2);
```

Set

```
HashSet<String> colors = new HashSet<>();
colors.add("red"); colors.add("white"); colors.add("red");
int two = colors.size();
```

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Data structures in java.util

Sequence

```
ArrayList<String> colors = new ArrayList<>();
colors.add("red"); colors.add("white"); colors.add("red");
String third = colors.get(2);
```

Set

```
HashSet<String> colors = new HashSet<>();
colors.add("red"); colors.add("white"); colors.add("red");
int two = colors.size();
```

Mapping

```
HashMap<String,String> colors = new HashMap<>();
colors.put("red", "piros"); colors.put("white", "fehér");
String whiteHu = colors.get("white");
```

Generic class

```
public class ArrayList<T> {
    public ArrayList() { ... }
    public T get(int index) { ... }
    public void set(int index, T item) { ... }
    public void add(T item) { ... }
    public T remove(int index) { ... }
    ...
}
```

Implementation of generic class

```
public class ArrayList<T> {
    private T[] data;
    private int size = 0;
    ...
    public T get(int index) {
        if (index < size) return data[index];
        else throw new IndexOutOfBoundsException();
    }
    ...
}
```

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Implementation of generic class

```
import java.util.Arrays;
public class ArrayList<T> {
    private T[] data;
    private int size = 0;
    ...
    public void add(T item) {
        if (size == data.length) {
            data = Arrays.copyOf(data,data.length+1);
        }
        data[size] = item;
        ++size;
    }
    ...
}
```

Allocation attempt: compilation error

```
public class ArrayList<T> {
    private T[] data;
    private int size = 0;
    ...
    public ArrayList() { this(256); }
    public ArrayList(int initialCapacity) {
        data = new T[initialCapacity];
    }
    ...
}
```

```
ArrayList.java:6: error: generic array creation
        data = new T[initialCapacity];
```

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Type erasure

- Type parameter: used during static type checking
- Target code: independent of type parameter
  - ◇ Haskell is similar
  - ◇ C++ *template* is different
- Compatibility with generic-less code
- Type parameter cannot be used run-time

The target code *can be considered* like this

```
public class ArrayList {
    private Object[] data;
    ...
    public ArrayList() { ... }
    public Object get(int index) { ... }
    public void set(int index, Object item) { ... }
    public void add(Object item) { ... }
    public Object remove(int index) { ... }
    ...
}
```

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Compatibility: raw type

```
import java.util.ArrayList;
...
ArrayList<String> parametrized = new ArrayList<>();
parametrized.add("Romeo");
parametrized.add(12);          // compilation error
String s = parametrized.get(0);
```

Compatibility: raw type

```
import java.util.ArrayList;
...
ArrayList<String> parametrized = new ArrayList<>();
parametrized.add("Romeo");
parametrized.add(12);        // compilation error
String s = parametrized.get(0);

ArrayList raw = new ArrayList();
raw.add("Romeo");
raw.add(12);
Object o = raw.get(0);
```

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Allocation: still not fixed

```
public class ArrayList<T> {
    private T[] data;
    private int size = 0;
    ...
    public ArrayList() { this(256); }
    public ArrayList(int initialCapacity) {
        data = new Object[initialCapacity];
    }
    ...
}
```

```
ArrayList.java:6: error: incompatible types:
                  Object[] cannot be converted to T[]
        data = new Object[initialCapacity];
                   ^
  where T is a type-variable:
```

Allocation – already valid

```
public class ArrayList<T> {
    private T[] data;
    private int size = 0;
    ...
    public ArrayList() { this(256); }
    public ArrayList(int initialCapacity) {
        data = (T[])new Object[initialCapacity];
    }
    ...
}
```

javac ArrayList.java

```
Note: ArrayList.java uses unchecked or unsafe operations.
Note: Recompile with -Xlint:unchecked for details.
```

Allocation – already valid, but still not perfect···

```
public class ArrayList<T> {
    private T[] data;
    private int size = 0;
    ...
    public ArrayList() { this(256); }
    public ArrayList(int initialCapacity) {
        data = (T[])new Object[initialCapacity];
    }
    ...
}
```

javac -Xlint:unchecked ArrayList.java

```
ArrayList.java:6: warning: [unchecked] unchecked cast
  required: T[]          found: Object[]
```

Casting somewhere else?

```
public class ArrayList<T> {
    private Object[] data;
    private int size = 0;
    ...
    public T get(int index) {
        if (index < size) return (T)data[index];
        else throw new IndexOutOfBoundsException();
    }
    ...
}
```

javac -Xlint:unchecked ArrayList.java

```
ArrayList.java:10: warning: [unchecked] unchecked cast
  required: T          found: Object
```

Warning-free

```
public class ArrayList<T> {
    private Object[] data;
    private int size = 0;
    ...
    @SuppressWarnings("unchecked")
    public T get(int index) {
        if (index < size) return (T)data[index];
        else throw new IndexOutOfBoundsException();
    }
    ...
}
```

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Inheritance

```
class A extends B { ... }
```

- A class is defined in terms of another
  - ◇ Only their difference is to be given: A Δ B
  - ◇ Reuse of code

Inheritance

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- Class `A` is the *child class* of `B`, the *parent class*

Inheritance

```
class A extends B { ... }
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- A class is defined in terms of another
  - ◇ Only their difference is to be given: A Δ B
  - ◇ Reuse of code
- Class `A` is the *child class* of `B`, the *parent class*
- Transitively:
  - ◇ subclass, derived class
  - ◇ super class, base class

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Inheritance

```
class A extends B { ... }
```

- A class is defined in terms of another
  - ◇ Only their difference is to be given: A Δ B
  - ◇ Reuse of code
- Class `A` is the *child class* of `B`, the *parent class*
- Transitively:
  - ◇ subclass, derived class
  - ◇ super class, base class
- No circularity!

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Example

```
public class Time {
  private int hour, min;     // initialized to 00:00
  public int getHour() { ... }
  public int getMin() { ... }
  public void setHour(int hour) { ... }
  public void setMin(int min) { ... }
  public void aMinPassed() { ... }
}
```

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Example

```
public class Time {
  private int hour, min;     // initialized to 00:00
  public int getHour() { ... }
  public int getMin() { ... }
  public void setHour(int hour) { ... }
  public void setMin(int min) { ... }
  public void aMinPassed() { ... }
}
```

```
public class ExactTime extends Time {
  private int sec;           // initialized to 00
  public int getSec() { ... }
  public void setSec(int sec) { ... }
  public boolean earlierThan(ExactTime that) { ... }
}
```

Implicit parent class

```
public class Time extends java.lang.Object {
  private int hour, min;     // initialized to 00:00
  public int getHour() { ... }
  public int getMin() { ... }
  public void setHour(int hour) { ... }
  public void setMin(int min) { ... }
  public void aMinPassed() { ... }
}
```

```
public class ExactTime extends Time {
  private int sec;           // initialized to 00
  public int getSec() { ... }
  public void setSec(int sec) { ... }
  public boolean earlierThan(ExactTime that) { ... }
}
```

java.lang.Object

Base class of every class!

```
package java.lang;
public class Object {
  public Object() { ... }

  public String toString() { ... }
  public int hashCode() { ... }
  public boolean equals(Object that) { ... }
  ...
}
```

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Constructors are not inherited

```
public class Time {
  private int hour, min;
  public Time(int hour, int min) {
    if (hour < 0 || hour > 23 || min < 0 || min > 59)
      throw new IllegalArgumentException();
    this.hour = hour;
    this.min = min;
  }
  ...
}
```

Constructors are not inherited

```
public class Time {
  private int hour, min;
  public Time(int hour, int min) {
    if (hour < 0 || hour > 23 || min < 0 || min > 59)
      throw new IllegalArgumentException();
    this.hour = hour;
    this.min = min;
  }
  ...
}
```

```
public class ExactTime extends Time {
  private int sec;
```

Constructors are not inherited

```
public class Time {
  private int hour, min;
  public Time(int hour, int min) {
    if (hour < 0 || hour > 23 || min < 0 || min > 59)
      throw new IllegalArgumentException();
    this.hour = hour;
    this.min = min;
  }
  ...
}
```

```
public class ExactTime extends Time {
  private int sec;
  public ExactTime(int hour, int min, int sec) { ? }
}
```

The child class needs a constructor!

```
public class Time {
  private int hour, min;
  public Time(int hour, int min) { ... }
  ...
}
```

```
public class ExactTime extends Time {
  private int sec;
  public ExactTime(int hour, int min, int sec) {
```

The child class needs a constructor!

```
public class Time {
  private int hour, min;
  public Time(int hour, int min) { ... }
  ...
}
```

```
public class ExactTime extends Time {
  private int sec;
  public ExactTime(int hour, int min, int sec) {
    super(hour, min); // must call parent's constructor
    if (sec < 0 || sec > 59)
      throw new IllegalArgumentException();
    this.sec = sec;
  }
}
```

Calling the `super(...)` constructor

- A constructor in the parent class
- For initializing inherited fields
- Must be the first statement!

Calling the `super(...)` constructor

- A constructor in the parent class
- For initializing inherited fields
- Must be the first statement!

Erroneous!!!

```
public class ExactTime extends Time {
  private int sec;
  public ExactTime(int hour, int min, int sec) {
    if (sec < 0 || sec > 59)
      throw new IllegalArgumentException();
    super(hour,min);
    this.sec = sec;
  }
}
```

Why is this correct? Missing **super**?!

```
public class Time extends Object {
  private int hour, min;
  public Time(int hour, int min) {
    if (hour < 0 || hour > 23 || min < 0 || min > 59)
      throw new IllegalArgumentException();
    this.hour = hour;
    this.min = min;
  }
  ...
}
```

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Implicit `super()` call

```
public class Time extends Object {
  private int hour, min;
  public Time(int hour, int min) {
    super();
    if (...) throw ...;
    this.hour = hour;
    this.min = min;
  }
  ...
}
```

```
package java.lang;
public class Object {
  public Object() { ... }
  ...
```

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Implicit parent class, implicit constructor, implicit super

```
class A {}
```

Implicit parent class, implicit constructor, implicit super

```
class A {}
```

```
class A extends java.lang.Object {
  A() {
    super();
  }
}
```

Constructors in a class

- One or more explicit constructors
- Default constructor

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Constructor body

1st statement

- Explicit `this` call
- Explicit `super` call
- Implicit (automatically generated) `super()` call (no-arg!)

Rest of the body

No calls using `this` or `super`!

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Interesting error

Seems OK, but···

```
class Base {
  Base(int n) {}
}

class Sub extends Base {}
```

Meaning

```
class Base extends Object {
  Base(int n) {
    super();
  }
}

class Sub extends Base {
  Sub() { super(); }
}
```

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A class defined with inheritance

- Members of parent class are inherited
- Can be extended with new members (Java: `extends`)
- Inherited instance methods can be redefined
  - ⋯and redeclared

Overriding instance methods

redefinition, overriding

```
package java.lang;
public class Object {
  ...
  public String toString() {...} //java.lang.Object@4f324b5c
}
```

Overriding instance methods

redefinition, overriding

```
package java.lang;
public class Object {
  ...
  public String toString() {...} //java.lang.Object@4f324b5c
}
```

```
public class Time {
  ...
  public String toString() {
    return hour + ":" + min; // 8:5
  }
}
```

Slightly better

redefinition, overriding

```
package java.lang;
public class Object {
  ...
  public String toString() {...} //java.lang.Object@4f324b5c
}
```

Slightly better

redefinition, overriding

```
package java.lang;
public class Object {
  ...
  public String toString() {...} //java.lang.Object@4f324b5c
}
```

```
public class Time {
  ...
  public String toString() {          // 8:05
    return "%1$d:%2$02d".formatted(hour, min);
  }
}
```

With the recommended @Override annotation

redefinition, overriding

```
package java.lang;
public class Object {
  ...
  public String toString() {...} //java.lang.Object@4f324b5c
}
```

With the recommended @Override annotation

redefinition, overriding

```
package java.lang;
public class Object {
  ...
  public String toString() {...} //java.lang.Object@4f324b5c
}
```

```
public class Time {
  ...
  @Override
  public String toString() {          // 8:05
    return "%1$d:%2$02d".formatted(hour, min);
  }
}
```

Calling super.toString()

```
package java.lang;                          // java.lang.Object@4f324b5c
public class Object {... public String toString() {...} ... }
```

Calling super.toString()

```
package java.lang;                    // java.lang.Object@4f324b5c
public class Object {... public String toString() {...} ... }

public class Time {
  ...
  @Override public String toString() {     // 8:05
    return "%1$d:%2$02d".formatted(hour, min);
  }
}
```

Calling super.toString()

```
package java.lang;                    // java.lang.Object@4f324b5c
public class Object {... public String toString() {...} ... }

public class Time {
  ...
  @Override public String toString() {    // 8:05
    return "%1$d:%2$02d".formatted(hour, min);
  }
}
public class ExactTime extends Time {
  ...
  @Override public String toString() {    // 8:05:17
    return "%1:%2$02d".formatted(super.toString(), sec);
  }
}
```

Overloading versus overriding

```
package java.lang;
public final class Integer extends Number {
   ...
   public static    int parseInt(String str)           { ... }
   ...
   public @Override String toString()                  { ... }
   public static    String toString(int i)             { ... }
   public static    String toString(int i, int radix) { ... }
   ...
}
```

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Differences

Overloading

- Methods/ctors with same name but different parameters
- Introduced method may overload inherited one (Java-specific)
- Compiler selects method/ctor based on actual parameters

Overriding

- Override a method defined in a base class
- Same name, same parameters
  - ◇ Same method
  - ◇ A method may have multiple implementations
- The most specific implementation is chosen run-time

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Static versus dynamic binding

System.out

```
public void println(   int value) { ... }
public void println(Object value) { ... } //value.toString()
...
```

```
System.out.println(7);                    // 7
System.out.println("Samurai");            // Samurai
System.out.println(new Time(21,30));      // 21:30
```

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`protected`: designed for inheritance

- Should be easy to subclass from
- Avoid accidental corruption of type invariant in subclasses

protected visibility

```
package java.util;
public abstract class AbstractList<E> implements List<E> {
   ...
   protected int modCount;
   protected AbstractList() { ... }
   protected void removeRange(int fromIndex, int toIndex) { ...
   ...
}
```

- In the same package
- In subclasses (potentially in different packages)

private $\subseteq$ package-private $\subseteq$ protected $\subseteq$ public

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No direct access to private components in subclasses!

```
class Counter {
  private int counter = 0;
  public int count() { return ++counter; }
}

class SophisticatedCounter extends Counter {
  public int count(int increment) {
    return counter += increment;   // compilation error
  }
}
```

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"Fixed"

```
class Counter {
  private int counter = 0;
  public int count() { return ++counter; }
}

class SophisticatedCounter extends Counter {
  public int count(int increment) {
    if (increment < 1) throw new IllegalArgumentException();
    while (increment > 1) {
      count();
      --increment;
    }
    return count();
  }
}
```

protected

```
package my.basic.types;
public class Counter {
  protected int counter = 0;
  public int count() { return ++counter; }
}
```

```
package my.advanced.types;
class SophisticatedCounter extends my.basic.types.Counter {
  public int count(int increment) {
    return counter += increment;
  }
}
```

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Abstraction: encapsulation and information hiding

```
public class Rational {
  private final int numerator, denominator;
  private static int gcd(int a, int b)              { ... }
  private void simplify()                           { ... }
  public Rational(int numerator, int denominator) { ... }
  public Rational(int value)             { this(value, 1); }
  public int getNumerator()              { return numerator; }
  public int getDenominator()            { return denominator; }
  public Rational times(Rational that) { ... }
  public Rational times(int that)      { ... }
  public Rational plus(Rational that)  { ... }
}
```

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Abstraction: encapsulation and information hiding

```
public class Rational {
  private final int numerator, denominator;
  private static int gcd(int a, int b)              { ... }
  private void simplify()                           { ... }
  public Rational(int numerator, int denominator) { ... }
  public Rational(int value)             { this(value, 1); }
  public int getNumerator()              { return numerator; }
  public int getDenominator()            { return denominator; }
  public Rational times(Rational that) { ... }
  public Rational times(int that)      { ... }
  public Rational plus(Rational that)  { ... }
}
```

- The interface of a class: everything `public` in it
  - ◇ Usually, all constructors and non-helper methods
    - ▶ Just the headers, e.g. `Rational times(Rational)`
  - ◇ Usually, none of the fields

An interface definition

```
public interface Rational {
  int getNumerator();
  int getDenominator();

  Rational times(Rational that);
  Rational times(int that);
  Rational plus(Rational that);
  ...
}
```

- interface members are `abstract` and `public` by default

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Contents of an `interface` definition

Declaration of instance methods: specification terminated with ;

```
int getNumerator();
```

Contents of an `interface` definition

Declaration of instance methods: specification terminated with ;

```
int getNumerator();
```

- Declaration of instance methods
  - ◇ with optional `default` implementation
- Definition of constants: `public static final`
- Static methods
- Nested (member) type definitions

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Generic interface

java/util/List.java

```
package java.util;
public interface List<T> {
    T get(int index);
    void set(int index, T item);
    void add(T item);
    ...
}
```

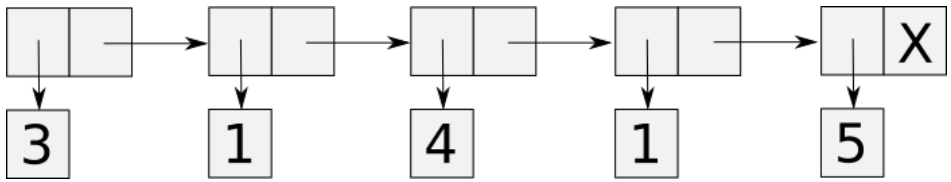
java/util/ArrayList.java

```
package java.util;
public class ArrayList<T> implements List<T> {
    public ArrayList() { ... }
    @Override public T get(int index) { ... }
    ...
```

Linked representation

```
package java.util;

public class LinkedList<T> implements List<T> {
    private T head;
    private LinkedList<T> tail;
    public LinkedList() { ... }
    @Override public T get(int index) { ... }
    @Override public void set(int index, T item) { ... }
    @Override public void add(T item) { ... }
    ...
}
```



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Implementation of an interface

Rational.java

```
public interface Rational {
    int getNumerator();
    int getDenominator();
    Rational times(Rational that);
}
```

Fraction.java

```
public class Fraction implements Rational {
  private final int numerator, denominator;
  public Fraction(int numerator, int denominator) { ... }
  public int getNumerator() { return numerator; }
  public int getDenominator() { return denominator; }
  public Rational times(Rational that) { ... }
}
```

Multiple implementations

Fraction.java

```
public class Fraction implements Rational {
  private final int numerator, denominator;
  public Fraction(int numerator, int denominator) { ... }
  public int getNumerator() { return numerator; }
  public int getDenominator() { return denominator; }
  public Rational times(Rational that) { ... }
}
```

Simplified.java

```
public class Simplified implements Rational {
  ...
  public int getNumerator() { ... }
  public int getDenominator() { ... }
  Rational times(Rational that) { ... }
```

Subtyping

```
class Fraction implements Rational { ... }
class ArrayList<T> implements List<T> { ... }
class LinkedList<T> implements List<T> { ... }
```

- Fraction <: Rational
- Simplified <: Rational
- For all T: ArrayList<T> <: List<T>
- For all T: LinkedList<T> <: List<T>

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Liskov's Substitution Principle

LSP: Liskov's Substitution Principle

Type A is the subtype of (base-)type B, if instances of A can be used wherever instances of B are used without a problem.

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An `interface` is a type

```
List<String> names;

static List<String> noDups(List<String> names) {
  List<String> retval = ...;
  ...
}
```

Not possible to instantiate

An `interface` cannot be instantiated

- No representation
- No constructor

```
List<String> names = new List<String>(); // compilation error
```

Not possible to instantiate

An `interface` cannot be instantiated

- No representation
- No constructor

```
List<String> names = new List<String>(); // compilation error
```

A `class` is a type which is allowed to be instantiated

```
ArrayList<String> names = new ArrayList<String>();
ArrayList<String> nicks = new ArrayList<>();
```

Not possible to instantiate

An `interface` cannot be instantiated

- No representation
- No constructor

```
List<String> names = new List<String>(); // compilation error
```

A `class` is a type which is allowed to be instantiated

```
ArrayList<String> names = new ArrayList<String>();
ArrayList<String> nicks = new ArrayList<>();
```

Good style: typing with an `interface`, instantiating with a `class`

```
List<String> names = new ArrayList<>();
```

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Static and dynamic type

Declared versus *actual* type of variable / parameter / ···

```
List<String> names = new ArrayList<>();


static List<String> noDups(List<String> names) {
    ... names ...
}


List<String> shortList = noDups(names);
```

abstract class

- Partially implemented classes
  - ◇ May contain `abstract` methods
- Cannot be instantiated
- Subclasses may be concrete

```
package java.util;
public abstract class AbstractList<E> implements List<E> {
  ...
  public abstract E get(int index);      // only declared
  public Iterator<E> iterator() { ... } // implemented
  ...
}
```

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Partial implementation

```
public abstract class AbstractCollection<E> ... {
  ...
  public abstract int size();
  public boolean isEmpty() {
    return size() == 0;
  }
  public abstract Iterator<E> iterator();
  public boolean contains(Object o) {
    Iterator<E> iterator = iterator();
    while (iterator.hasNext()) {
      E e = iterator.next();
      if (o==null ? e==null : o.equals(e)) return true;
    }
    return false;
  }
}
```

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Concrete subclass

```
public abstract class AbstractCollection<E> implements Collection<E>
    ...
    public abstract int size();
}
```

```
public abstract class AbstractList<E> extends AbstractCollection<E>
                                      implements List<E>
  ...
  public abstract E get(int index);
}
```

```
public class ArrayList<E> extends AbstractList<E> {
  ...
  public int size() { ... }              // implemented
  public E get(int index) { ... }      // implemented
}
```

Multiple inheritance

- A type inherits from multiple types directly
- In Java: multiple interfaces
- MI raises problems

Examples

OK

```
package java.util;
public class Scanner implements Closeable, Iterator<String>
{ ... }
```

OK

```
interface PoliceCar extends Car, Emergency { ... }
```

Erroneous

```
class PoliceCar extends Car, Emergency { ... }
```

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Hypothetically

```
class Base1 {
    int x;
    void setX(int x) { this.x = x; }
    ...
}

class Base2 {
    int x;
    void setX(int x) { this.x = x; }
    ...
}

class Sub extends Base1, Base2 { ... }
```

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Hypothetically: diamond-shaped inheritance

```
class Base0 {
    int x;
    void setX(int x) { this.x = x; }
    ...
}

class Base1 extends Base0 { ... }

class Base2 extends Base0 { ... }

class Sub extends Base1, Base2 { ... }
```

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Differences between classes and interfaces

- Classes can be instantiated
  - ◇ `abstract class`?
- Classes support only single inheritence
  - ◇ `final class`?
- Classes may contain instance fields
  - ◇ In interfaces: `public static final`

Multiple inheritance from interfaces

```
interface Base1 {
    abstract void setX(int x);
    ...
}

interface Base2 {
    abstract void setX(int x);
    ...
}

class Sub implements Base1, Base2 {
    void setX(int x) { ... }
    ...
}
```

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Two aspects of inheritance

- Code reuse
- Subtyping

Inheritance gives rise to subtyping

$$A \ \Delta \ B \ \Rightarrow \ A <: B$$

Inheritance gives rise to subtyping

$$A \ \Delta \ B \ \Rightarrow \ A <: B$$

```
public class ExactTime extends Time { ... }
```

- `ExactTime` has everything `Time` has
- Whatever you can do with `Time`, you can do with `ExactTime`
- `ExactTime` $<:$ `Time`

Inheritance gives rise to subtyping

$$A\ \Delta\ B\ \Rightarrow\ A <: B$$

```
public class ExactTime extends Time { ... }
```

- `ExactTime` has everything `Time` has
- Whatever you can do with `Time`, you can do with `ExactTime`
- `ExactTime` $<:$ `Time`
- $\forall T$ class : $T <:$ `java.lang.Object`

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Subtyping

```
public class Time {
    ...
    public void aMinutePassed() { ... }
    public boolean sameHourAs(Time that) { ... }
}
```

```
public class ExactTime extends Time {
    ...
    public boolean isEarlierThan(ExactTime that) { ... }
}
```

```
ExactTime time = new ExactTime();          // 0:00:00
time.aMinutePassed();                      // 0:01:00
time.sameHourAs(new ExactTime())           // true
```

Polymorphic references

```
public class Time {
  ...
  public void aMinutePassed() { ... }
  public boolean sameHourAs(Time that) { ... }
}
```

```
public class ExactTime extends Time {
  ...
  public boolean isEarlierThan(ExactTime that) { ... }
}

ExactTime time1 = new ExactTime();
Time      time2 = new ExactTime(); // upcast
time2.sameHourAs(time1)
```

Static and dynamic type

Static type: *declared* type of variable / parameter / return value / ···

- Follows from program text
- Does not change during program execution
- Is used by the compiler during static type checking

```
Time time
```

Static and dynamic type

Static type: *declared* type of variable / parameter / return value / ···

- Follows from program text
- Does not change during program execution
- Is used by the compiler during static type checking

```
Time time
```

Dynamic type: *actual* type of variable / parameter / return value / ···

- May change during program execution
- Is meaningful only during run time
- Is the subtype of the static type

```
time = ... ? new ExactTime() : new Time()
```

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Overriding

```
package java.lang;                          // java.lang.Object@4f324b5c
public class Object {... public String toString() {...} ... }
```

Overriding

```
package java.lang;                          // java.lang.Object@4f324b5c
public class Object {... public String toString() {...} ... }

public class Time {
  ...
  @Override public String toString() {     // 8:05
    return "%1$d:%2$02d".formatted(hour, min);
  }
}
```

Overriding

```
package java.lang;                          // java.lang.Object@4f324b5c
public class Object {... public String toString() {...} ... }

public class Time {
  ...
  @Override public String toString() {     // 8:05
    return "%1$d:%2$02d".formatted(hour, min);
  }
}
public class ExactTime extends Time {
  ...
  @Override public String toString() {     // 8:05:17
    return "%1:%2$02d".formatted(super.toString(), sec);
  }
}
```

Overloading versus overriding

Overloading

- Same name, different formal parameters
- Inherited methods can be overloaded (in Java!)
- Compiler selects method based on actual parameters

Overriding

- An instance method defined in a base class
- Same name, same formal parameters (same signature)
  - ◇ Same method
  - ◇ Multiple implementations
- **The "most specific" implementation is selected at run time**

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Dynamic binding (or *late binding*)

```
ExactTime e = new ExactTime();
Time      t = e;
Object    o = t;

System.out.println(e.toString());      // 0:00:00
System.out.println(t.toString());      // 0:00:00
System.out.println(o.toString());      // 0:00:00
```

At the invocation of an instance method, the implementation that matches the dynamic type of the privileged parameter best will be selected.

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The role of static and dynamic type

Static type

What can we do with an expression?

- Static type checking

```
Object o = new Time();
o.setHour(8);                    // compilation error
```

Dynamic type

- Which implementation of an instance method?

```
Object o = new Time();
System.out.println(o);   // select toString()
                         // implementation
```

- Dynamic type checking

Inheritance example

```
package company.hr;
public class Employee {
  String name;
  int basicSalary;
  java.time.ZonedDateTime startDate;
  ...
}
```

Inheritance example

```
package company.hr;
public class Employee {
  String name;
  int basicSalary;
  java.time.ZonedDateTime startDate;
  ...
}
```

```
package company.hr;
import java.util.*;
public class Manager extends Employee {
  final HashSet<Employee> workers = new HashSet<>();
  ...
}
```

Parent class

```
package company.hr;
import java.time.ZonedDateTime;
import static java.time.temporal.ChronoUnit.YEARS;
public class Employee {
  ...
  private ZonedDateTime startDate;
  public int yearsInService() {
    return (int) startDate.until(ZonedDateTime.now(), YEARS);
  }
  private static int bonusPerYearInService = 0;
  public int bonus() {
    return yearsInService() * bonusPerYearInService;
  }
}
```

Child class

```
package company.hr;
import java.util.*;
public class Manager extends Employee {
  // inherited: startDate, yearsInService() ...
  ...
  private final Set<Employee> workers = new HashSet<>();
  public void addWorker(Employee worker) {
    workers.add(worker);
  }
  private static int bonusPerWorker = 0;
  @Override public int bonus() {
    return workers.size() * bonusPerWorker + super.bonus();
  }
}
```

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Dynamic binding also in inherited methods

```
public class Employee {
  ...
  private int basicSalary;
  public int bonus() {
    return yearsInService() * bonusPerYearInService;
  }
  public int salary() { return basicSalary + bonus(); }
}
```

```
public class Manager extends Employee {
  ...
  @Override public int bonus() {
    return workers.size()*bonusPerWorker + super.bonus();
  }
}
```

Dynamic binding also in inherited methods

```
Employee jack = new Employee("Jack", 10000);
Employee pete = new Employee("Pete", 12000);
Manager eve = new Manager("Eve", 12000);
Manager joe = new Manager("Joe", 12000);
eve.addWorker(jack);
joe.addWorker(eve);      // polymorphic formal parameter
joe.addWorker(pete);

Employee[] company = {joe, eve, jack, pete};//<-heterogeneous
                                            // data structure
int totalSalaryCosts = 0;
for (Employee e: company) {
  totalSalaryCosts += e.salary();
}
```

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Dynamic binding

At the invocation of an instance method, the implementation that matches the dynamic type of the privileged parameter best will be selected.

Fields and static methods cannot be overridden

```
class Base {
  int field = 3;
  int iMethod() { return field; }
  static int sMethod() { return 3; }
}

class Sub extends Base {
  int field = 33;                          // hiding
  static int sMethod() { return 33; } // hiding
}
```

```
Sub sub = new Sub();          Base base = sub;
sub.sMethod()  ==  33         base.sMethod()  ==  3
sub.field      ==  33         base.field      ==  3
sub.iMethod()  ==  3          base.iMethod()  ==  3
```

Type conversions between primitive types

Automatic type conversion (transitive)

- `byte` → `short` → `int` → `long`
- `long` → `float`
- `float` → `double`
- `char` → `int`
- `byte b = 42;` and `short s = 42;` and `char c = 42;`

Explicit type cast

```
int i = 42;
short s = (short)i;
```

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Wrapper classes

Implicitly imported (`java.lang`), immutable classes

- `java.lang.Boolean` – `boolean`
- `java.lang.Character` – `char`
- `java.lang.Byte` – `byte`
- `java.lang.Short` – `short`
- `java.lang.Integer` – `int`
- `java.lang.Long` – `long`
- `java.lang.Float` – `float`
- `java.lang.Double` – `double`

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The interface of java.lang.Integer (fragment)

```
static int MAX_VALUE     // 2^31-1
static int MIN_VALUE     // -2^31

static int compare(int x, int y)     // 3-way comparison
static int max(int x, int y)
static int min(int x, int y)
static int parseInt(String str [, int radix])
static String toString(int i [, int radix])
static Integer valueOf(int i)

int compareTo(Integer that)          // 3-way comparison
int intValue()
```

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Auto-(un)boxing

- Automatic two-way conversion
- Between primitive type and its wrapper class

```
Integer ref = 42;
int pri = ref;

Integer sum = ref + pri;
```

```
Integer ref = Integer.valueOf(42);
int     pri = ref.intValue();
Integer sum = Integer.valueOf(
                  ref.intValue()
                  + pri          );
```

Auto-(un)boxing

- Automatic two-way conversion
- Between primitive type and its wrapper class

```
Integer ref = 42;
int pri = ref;

Integer sum = ref + pri;
```

```
Integer ref = Integer.valueOf(42);
int     pri = ref.intValue();
Integer sum = Integer.valueOf(
                  ref.intValue()
                  + pri          );
```

```
ArrayList<Integer> numbers = new ArrayList<>();
numbers.add(7);
int seven = numbers.get(0);
```

Puzzle 3: Long Division (Bloch & Gafter: Java Puzzlers)

```
public class LongDivision {
  public static void main(String[] args) {
    final long MICROS_PER_DAY = 24 * 60 * 60 * 1000 * 1000;
    final long MILLIS_PER_DAY = 24 * 60 * 60 * 1000;
    System.out.println(MICROS_PER_DAY / MILLIS_PER_DAY);
  }
}
```

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Unboxing with `null`

Boxing always works

```
int val = 42;
Integer value = val;
```

Unboxing with `null`

Boxing always works

```
int val = 42;
Integer value = val;
```

Unboxing *almost* always works

```
Integer value = 42;
int val = value;
```

Unboxing with `null`

Boxing always works

```
int val = 42;
Integer value = val;
```

Unboxing *almost* always works

```
Integer value = 42;
int val = value;
```

…except when it doesn't

```
Integer value = null;
int val = value;        // NullPointerException
```

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Auto-(un)boxing costs more

```
int n = 10;
int fact = 1;
while (n > 1) {
    fact *= n;
    --n;
}
```

```
Integer n = 10;
Integer fact = 1;
while (n > 1) {
    fact *= n;
    --n;
}
```

Auto-(un)boxing costs more

```
int n = 10;
int fact = 1;
while (n > 1) {
    fact *= n;
    --n;
}
```

```
Integer n = 10;
Integer fact = 1;
while (n > 1) {
    fact *= n;
    --n;
}
```

Meaning:

```
Integer n = Integer.valueOf(10);
Integer fact = Integer.valueOf(1);
while (n.intValue() > 1) {
    fact = Integer.valueOf(fact.intValue() * n.intValue());
    n = Integer.valueOf(n.intValue() - 1);
}
```

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Conversions on reference types

- Automatic (upcast) – subtyping
- Explicit (downcast) – type-cast operator

Conversions on reference types

- Automatic (upcast) – subtyping
- Explicit (downcast) – type-cast operator

Inheritance – subtyping: `class A extends B` ⋯

- $A <: B$
- $\forall T: \quad T <:$ `java.lang.Object`

Type cast (downcast)

- The static type of the expression "`(Time)o`" is `Time`

Type cast (downcast)

- The static type of the expression "`(Time)o`" is `Time`
- If the dynamic type of o is `Time`:

```
Object o = new Time(3,20);
o.aMinPassed();              // compilation error
((Time)o).aMinPassed();      // compiles. works.
```

Type cast (downcast)

- The static type of the expression "`(Time)o`" is `Time`
- If the dynamic type of o is `Time`:

```
Object o = new Time(3,20);
o.aMinPassed();              // compilation error
((Time)o).aMinPassed();      // compiles. works.
```

- If not, `ClassCastException` is thrown

```
Object o = "Twenty past three";
o.aMinPassed();              // compilation error
((Time)o).aMinPassed();      // run-time error
```

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Dynamic type checking

- During run-time, based on dynamic type
- More precise than static type checking
  - ◇ Dynamic types can be subtypes
- Flexibility
- Safety: only when explicitly requested (type cast)

instanceof operator

```
Object o = new ExactTime(3,20,0);
...
if (o instanceof Time t) {
    t.aMinPassed();
}
```

- Dynamic type of given expression is a subtype of given type

instanceof operator

```
Object o = new ExactTime(3,20,0);
...
if (o instanceof Time t) {
    t.aMinPassed();
}
```

- Dynamic type of given expression is a subtype of given type
- Static type and given type have to be related

```
"apple" instanceof Integer    // compilation error
```

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instanceof operator

```
Object o = new ExactTime(3,20,0);
...
if (o instanceof Time t) {
    t.aMinPassed();
}
```

- Dynamic type of given expression is a subtype of given type
- Static type and given type have to be related

```
"apple" instanceof Integer    // compilation error
```

- `null` yields `false`

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Representation of dynamic types during runtime

- objects of class `java.lang.Class`
- can be accessed run-time

```
Object o  = new Time(17,25);
Class  c  = o.getClass();      // Time.class
Class  cc = c.getClass();      // Class.class
```

A value belongs to a type (subtyping allowed)

```
String str = "Java";
Object o = str;
Integer i = (o instanceof Integer) ? (Integer)o : null;
```

Exact match of dynamic types

```
Integer i = o.getClass().equals(Integer.class) ?
                (Integer)o : null;
```